

B5-BEYOND-LAYER-BOUND claim-level synthesis: the beyond-layer machinery (DR-2, STEP-5B, SC-SCOPE)

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

Claim-level synthesis for B5-BEYOND-LAYER-BOUND. It adds no mathematics; every statement carries the unit's registered tier. **Scope:** the composition of B5's units into the canonical claim state, and the precise tier relations between the claim (now T6-conditional) and its units. **Not claimed:** the vacuum selection itself (that is B1/B2's head), arbitrary non-lattice DR-2, or full STEP-5B exhaustiveness beyond the amended-class pin.

2. The claim and its canonical state

Claim (canonical, post-T-031). Pattern-generic Gershgorin reduction of the beyond-layer bound: T6 PROVED-CONDITIONAL on $\mathcal{H}_{B5}^{T6} = \{\text{lattice-class H-ADM-COH discharged; non-lattice remainder excluded (T-030); SC-SCOPE endpoint thinness accepted; sunset-limited joint } \times 1.040 > 1 \text{ accepted}\}$; scope CLOSED@LATTICE-DISCHARGED + NONLATTICE-REMAINDER; label B5-BeyondLayer-T6Conditional-260612. (Historical: T5 PINNED-CLOSURE from 2026-06-06 until the T-031 verdicts of 2026-06-12.)

3. Sub-proof unit map (per-unit tier and role)

Unit	Tier	What it contributes	Publication state
DR-2/	T7 (lattice, std. NT)	Lemma 1 backing: $E_+(Q) \leq_\epsilon R^\epsilon N^2$ decoupling-free on lattice shells; Lemma NT pinned in-bundle	PUBLISHED – DR2-Lattice-T7-NTstandard-260612 (8/8, 38/38)
STEP-5B/	T6	the rectangle prefactor $K(n) \leq 8 + 4\sqrt{14}\kappa^4\sqrt{n}$ + official threshold $n_{\text{adm}} \lesssim 1.59 \times 10^5$	PUBLISHED – STEP-5B-Rectangle-T6-260612 (192/192)
SC-SCOPE/	T5 thin	the (S) selection floor at the endpoint: joint = $\times 1.040 \times 1.082 > 1$ on $\mathcal{W}_{SC}$	PUBLISHED – SC-SCOPE-T5-260612 (4/4, 29/29)
H-LAYER-AUX/	T5	RES-4 layer-ratio positivity $\rho(I) \geq \times 2.58 > 1$, monotone (201 nodes) – input to STEP-5B's K-budget	AUXILIARY (operator-accepted v1.1)
T5-DOSSIER/	T5	the assignment dossier pinning the claim tier (closure pillars, 192/192)	internal consolidation

4. Composition: the claim tier vs its units (updated post-T-031)

The claim's statement is the class-scoped closure; its strongest units exceed it (DR-2 T7-NTstandard; STEP-5B rectangle T6; SC-SCOPE T5 thin) and are cited at their own tiers. The upgrade path foreseen here (the T-031 exhaustiveness decision layer + the R-U6-1 backlog lemma) was EXECUTED on 2026-06-12: R-U6-1/2 discharged, T-031 verdicts D1-A/D2-A/D3-A enacted, and the claim re-tiered to T6 PROVED-CONDITIONAL on the explicit set $\mathcal{H}_{B5}^{T6}$ (the budget comparison was already machine-closed

at $\times 55.6 / \times 8.8 / \times 2.1\text{--}2.6$). The remaining T7 path: remove the two acceptance hypotheses – harden the sunset endpoint beyond thin, and close T-030.

Consumption map (who uses what): B1/B2’s head uses Lemma 1 (DR-2 mechanism) + Lemma 2 (in-head packing) for (O), SC-SCOPE for (S), STEP-5B for the K-budget; H-LAYER-AUX feeds STEP-5B. Per-edge accounting: parent capstone Main-Line-Synthesis-T013-260612 Sec. 3–4.

5. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) “*A claim holding a higher-tier unit invites tier-shopping: citing the claim at its unit’s tier.*” ADRESSED: Sec. 4 fixes the rule – cite the CLAIM at its own tier (now T6-conditional), the UNIT at its bundle tier (its own statement). The two statements differ; the synthesis exists precisely to prevent the conflation.
- (β) “*SC-SCOPE thinness ($\times 1.04$) could silently cap the whole B5 machinery.*” VALID with mitigation: the thinness affects only the (S) endpoint; the off-diagonal chain (DR-2 + STEP-5B + RES-4) carries $\geq \times 2.1$ margins everywhere on the band, and the head package consumes (S) with the thinness explicitly flagged (parent capstone Sec. 4a).
- (γ) “*The amended-class pin (H-ADM-COH) is an operator decision – is the claim then ‘proved’?*” VALID and intrinsic: T5 PINNED-CLOSURE is the honest label for exactly this situation (closure WITHIN an operator-pinned class definition); the exhaustiveness question is tracked as T-031, not hidden.

Quantitative sanity check (mandatory). Cross-unit consistency: the STEP-5B rectangle bound at the class size ($K(n_{\text{pack}}) \leq 113$) vs the measured $K = 107$ (6%, safe side); RES-4’s $\rho \geq 2.58$ vs the sharpened floor $\rho_{\text{lat}} = 6.55$ (consistent: RES-4 predates the $K \leq T'$ sharpening); SC-SCOPE joint $\times 1.040$ reproduced from its terms $[(1 + 0.385)/6.55 + (1 - 1/6.55)/1.13]^{-1}$; all three bundle digests verified at publication.

6. Result footer

Result ID:	B5-BEYOND-LAYER-BOUND-Synthesis-T013-260612 (PUBLISHED claim-level SYNTHESIS, T-013)
Precise statement:	B5’s canonical state (post-T-031: T6 PROVED-CONDITIONAL on H_B5^T6, label B5-BeyondLayer-T6Conditional-260612) is the claim’s own statement; its sub-proof units hold PUBLISHED bundles at their own tiers (DR-2 T7-lattice, STEP-5B T6, SC-SCOPE T5 thin) and are cited at those tiers by the B1/B2 head. Claim tier != max unit tier.
Scope:	synthesis layer; no new math, no tier change, no gate flip.
Evidence grade:	SYNTHESIS over three PUBLISHED bundles + operator-accepted auxiliaries.
Reproduction command:	run the three bundles’ entry manifests (8 + 1 + 4 scripts).
Falsifier:	any falsifier of the cited footers; or a unit-tier mismatch against INDEX.md.
Tier before / after:	no change BY THIS NOTE; the claim itself was re-tiered T5 -> T6-conditional by T-031 (D3-A, 2026-06-12), reflected in this v1.2.
No-overclaim statement:	the claim is cited at T6-conditional; unit tiers are not claim tiers; T7 requires removing the two acceptance hypotheses

(sunset hardening; T-030).
Next required action: sunset endpoint hardening (T7 path) and
T-030 frontier, per the active tracks.